EE2004 Digital Signal Processing

Tutorial 10 Answer Key

1. Nyquist rate in Hz: (a) 20, (b) 200, (c) 50, (d) 70, (e) not bandlimited
2. (a) No aliasing, (b) yes, there will be aliasing.
3. (a) $\omega_0 = \frac{2\pi}{3}$, (b) $\theta = -\phi$
4. 250π, 2250π
5. $7 \sin\left(\frac{2\pi n}{5}\right) - 3 \cos\left(\frac{12\pi n}{25}\right)$
6. $F_s = 160\,\text{Hz}$
7. Plots
8. (a) $2\Omega_c$, (b) $\frac{\Omega_c}{5}$, (c) $4\Omega_c$, (d) Ω_c, (e) Ω_c
9. (a) $T = \frac{1}{100}$
   (b) Hint: Not unique. Find another T that is consistent with the given information.